Service Bulletin

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FleetguardFIT™ Filtration Monitoring System Troubleshooting	

FleetguardFIT™ Filtration Monitoring System Troubleshooting

Purpose:

Troubleshooting of FleetguardFIT™ filtration monitoring system.

- FleetguardFIT™ system not working as expected.
- FleetguardFIT™ data not visible on telematics portal.
- FleetguardFIT™ power issue.
- FleetguardFIT™ data incorrect on portal.

This symptom tree can be used to troubleshoot the FleetguardFIT™ filtration monitoring system. Perform the list of troubleshooting steps in the sequence shown.

Troubleshooting Summary

STEPS	
STEP 1. Identify telematics portal state	
	STEP 1A. Check for faulty equipment visibility on telematics portal
	STEP 1B. Check for visibility of other equipment on telematics portal
STEP 2. Confirm telematics device operating properly	
	STEP 2A. Telematics device power
	STEP 2B. Telematics device data communication
STEP 3. Identify state of FleetguardFIT™ Filter Monitor System (FMS) LEDs	
	STEP 3A. LED operation
	STEP 3B. Blue LED steady on
	STEP 3C. Green and red LED flashing
STEP 4. Identify state of power supply	
	STEP 4A. Fuse condition
	STEP 4B. Power supply, ignition, and ground wire condition

STEPS	
	STEP 4C. Power supply, ignition, and chassis ground connections
	STEP 4D. Datalink harness connection to FleetguardFIT™ FMS
STEP 5. Equipment J1939 connections	
	STEP 5A. Inspect equipment and FleetguardFIT™ J1939 connection
	STEP 5B. Inspect equipment and telematics J1939 connection
	STEP 5C. Equipment J1939 public connection
	STEP 5D. Measure equipment resistance
	STEP 5E. Inspect J1939 backbone terminal resistors
STEP 6. Identify FleetguardFIT™ FMS compatibility	
	STEP 6A. Communication baud rate
STEP 7. Identify state of FleetguardFIT™ sensor(s)	
	STEP 7A. Check telematics portal for FleetguardFIT™ oil quality sensor data
	STEP 7B. Oil quality sensor data
	STEP 7C. Check telematics portal for FleetguardFIT™ differential pressure and restriction sensor data
	STEP 7D. differential pressure and restriction sensor data
STEP 8. Identify state of FleetguardFIT™ sensor installations	
	STEP 8A. Oil quality sensor
	STEP 8B. Air restriction sensor
	STEP 8C. differential pressure sensor hardware
STEP 9. Identify state of FleetguardFIT™ sensor extension harnesses and FMS	
	STEP 9A. Sensor breakout harness and extension harness condition
	STEP 9B. Sensor breakout harness and extension harness connection
	STEP 9C. Sensor breakout harness and extension harness continuity
	STEP 9D. FMS module replacement

STEP 1. Identify telematics portal state.

STEP 1A. Check for faulty equipment visibility on telematics portal.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Log on to telematics portal and check for operation	Is equipment in question visible on portal? **Yes**	**3A**
	Is equipment in question visible on portal? **No**	**1B**

STEP 1B. Check for visibility of other equipment on telematics portal.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Check telematics portal for visibility of other equipment	Is other equipment visible on portal? **Yes** **Note:** Not an issue with portal	**2A**
	Is other equipment visible on portal? **No**	**Contact:** Telematics service provider support

STEP 2. Confirm telematics device operating properly.

STEP 2A. Telematics device power.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Check telematics device power supply	Does the telematics device have power? **Yes** **Note:** Not an issue with telematics device power supply	**2B**
	Does the telematics device have power? **No**	**Contact:** Telematics service provider support

STEP 2B. Telematics device data communication.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Check telematics device for data communication	Is the telematics device sending / receiving data? **Yes** **Note:** Not an issue with telematics device data communication	**3A**
	Is the telematics device sending / receiving data? **No**	**Contact:** Telematics service provider support **Note:** May be issue with SIM card, data limits, internal device issue, device antenna, cell service, or other

STEP 3. Identify state of FleetguardFIT™ FMS LEDs.

STEP 3A. LED operation.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Inspect FMS module	Are any LEDs on? **Yes** **Note:** FMS is receiving power	**3B**
	Are any LEDs on? **No**	**4A**

STEP 3B. Blue LED steady on.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Inspect FMS module	Is blue LED steady on? **Yes** **Note:** Ignition power source is wired correctly	**3C**
	Is blue LED steady on? **No** **Note:** Ignition power source needs to be connected or repaired	**4A**

STEP 3C. Green and red LED flashing.

<table>
<tr><th colspan="3">Conditions : While equipment is on</th></tr>
<tr><th>Action</th><th>Specification/Repair</th><th>Next Step</th></tr>
<tr><td rowspan="2">Inspect FMS module</td><td>Are green and red LEDs flashing

Yes

Note:
FMS is sending / receiving data</td><td>5B</td></tr>
<tr><td>Are green and red LEDs flashing

No</td><td>5A</td></tr>
</table>

STEP 4. Identify state of power supply.

STEP 4A. Fuse condition.

<table>
<tr><th colspan="3">Conditions : While equipment is off</th></tr>
<tr><th>Action</th><th>Specification/Repair</th><th>Next Step</th></tr>
<tr><td rowspan="2">Inspect inline fuses for ignition and power supply</td><td>Is fuse open?

Yes

Repair:
Replace 10 ampere fuse</td><td>Repair Complete</td></tr>
<tr><td>Is fuse open?

No</td><td>4B</td></tr>
</table>

STEP 4B. Power supply, ignition, and ground wire condition.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Carefully inspect datalink harness	Are power supply, ignition, and ground wires in good condition with no damage, i.e. no tears? **Yes**	**4C**
	Are power supply, ignition, and chassis ground wires in good condition with no damage? **No** **Repair:** Replace or replace datalink harness	**Repair Complete**

STEP 4C. Power supply, ignition, and chassis ground connections.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Carefully inspect datalink harness	Are power supply, ignition, and ground wires connected? **Yes**	**4D**
	Are power supply, ignition, and chassis ground wires connected? **No** **Repair:** Connect wire(s) that are not connected	**Repair Complete**

STEP 4D. Datalink harness connection to FleetguardFIT™ FMS.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Inspect datalink harness connection to FMS module	Is the datalink harness connected securely to FMS module? **Yes**	**5A**
	Is the datalink harness connected securely to FMS module? **No** **Repair:** Connect datalink harness to FMS module	**Repair Complete**

STEP 5. Equipment J1939 connections.

STEP 5A. Inspect equipment and FleetguardFIT™ J1939 connection.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Inspect FleetguardFIT™ J1939 connection with equipment	Is FleetguardFIT™ properly connected to equipment's J1939 public data connection? **Yes**	**5B**
	Is FleetguardFIT™ properly connected to equipment's J1939 public data connection? **No** **Repair:** Connect FleetguardFIT™ properly to J1939 public datalink	**Repair Complete**

STEP 5B. Inspect equipment and telematics J1939 connection.

<table>
<tr><td colspan="3">Conditions : While equipment is off</td></tr>
<tr><td>Action</td><td>Specification/Repair</td><td>Next Step</td></tr>
<tr><td rowspan="2">Inspect telematics J1939 connection with equipment</td><td>Is telematics device properly connected to equipment’s J1939 public data connection?

Note:
FleetguardFIT™ FMS and telematics should be on different nodes of the datalink backbone

Yes</td><td>5C</td></tr>
<tr><td>Is telematics device properly connected to equipment’s J1939 public data connection?

No

Repair:
Connect telematics properly to J1939 public datalink</td><td>Repair Complete</td></tr>
</table>

STEP 5C. Equipment J1939 public connection.

<table>
<tr><td colspan="3">Conditions : While equipment is off</td></tr>
<tr><td>Action</td><td>Specification/Repair</td><td>Next Step</td></tr>
<tr><td rowspan="2">Measure continuity from 9-pin service connector to 3-pin Deutsch connector

Note:
Needs to be completed for FleetguardFIT™ and telematics J1939 connections</td><td>Was continuity confirmed?

Yes</td><td>5D</td></tr>
<tr><td>Was continuity confirmed?

No

Repair:
Locate different public J1939 public connection or repair wiring if 3 pin is known to be a public J1939 connection</td><td>Repeat Step</td></tr>
</table>

STEP 5D. Measure equipment resistance.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Disconnect the J1939 connection; using the equipment J1939 3 pin or 9 Pin service connector measure the equipment resistance	Is the measured resistance 55-65 Ohms? **Yes**	**6A**
	Is the measured resistance 55-65 Ohms? **No**	**5E**

STEP 5E. Inspect J1939 backbone terminal resistors.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Inspect J1939 connection terminal resistors; measure resistance Note: ONLY two 120 Ohms resistors needed in parallel on backbone to achieve a total resistance of 60 Ohms	Is the measured resistance in each resistor 120 Ohms? **Yes**	**6A**
	Is the measured resistance in each resistor 120 Ohms? **No** **Repair:** Discard and replace resistor(s)	**Repair Complete**

STEP 6. Identify FleetguardFIT™ FMS compatibility.

STEP 6A. Communication baud rate.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Determine engine, Telematics Service Provider(TSP), and FleetguardFIT™ FMS baud rate **Note:** FleetguardFIT™ FMS modules options are 250 kbps and 500 kbps	Do engine, TSP, and FleetguardFIT™ FMS baud rates match? **Yes**	**7A**
	Do engine, TSP, and FleetguardFIT™ FMS baud rates match? **No** **Repair:** Replace with proper FleetguardFIT™ FMS module	**Repair Complete**

STEP 7. Identify state of FleetguardFIT™ sensor(s).

STEP 7A. Check telematics portal for FleetguardFIT™ oil quality sensor data.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Log on to telematics portal and check for FleetguardFIT™ oil quality sensor data	Is any oil quality sensor data visible on portal? **Yes**	**7B**
	Is any oil quality sensor data visible on portal? **No**	**9A**

STEP 7B. Oil quality sensor data.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Log on to telematics portal and check for FleetguardFIT™ oil quality sensor data	Is oil quality sensor data within expected ranges? **Yes**	**7C**
	Is oil quality sensor data within expected ranges? **No**	**8A**

STEP 7C. Check telematics portal for FleetguardFIT™ differential pressure and restriction sensor data.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Log on to telematics portal and check for FleetguardFIT™ differential pressure and restriction sensor data	Is any differential pressure and restriction sensor data visible on portal? **Yes**	**7D**
	Is any differential pressure and restriction sensor data visible on portal? **No**	**9A**

STEP 7D. differential pressure and restriction sensor data.

Conditions : While equipment is on		
Action	**Specification/Repair**	**Next Step**
Log on to telematics portal and check for FleetguardFIT™ differential pressure and restriction sensor data	Is differential pressure and restriction sensor data within expected ranges? **Yes**	**Repair Complete**
	Is differential pressure and restriction sensor data within expected ranges? **No**	**8B**

STEP 8. Identify state of FleetguardFIT™ sensor installations.

STEP 8A. Oil quality sensor.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Inspect oil quality sensor installation	Is the sensor tip located in an area with hot pressurized flowing oil and no leaks? **Yes**	**9A**
	Is the sensor tip located in an area with hot pressurized flowing oil and no leaks? **No** **Repair:** Relocate oil quality sensor according to FleetguardFIT™ instructions	**Repair Complete**

STEP 8B. Air restriction sensor.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Inspect air restriction sensor installation	Is the sensor visibly in good condition and connected to air intake system with no leaks? **Yes**	**8C**
	Is the sensor visibly in good condition and connected to air intake system with no leaks? **No** **Repair:** Replace restriction sensor	**Repair Complete**

STEP 8C. differential pressure sensor(s).

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Inspect differential pressure sensor installation	Is the sensor properly connected to an inlet and outlet port of the filter head with no leaks? **Yes**	**9A**
	Is the sensor properly connected to an inlet and outlet port of the filter head with no leaks? **No** **Repair:** Properly connect differential pressure sensor to inlet/outlet ports of the filter head. Replace hardware if needed.	**Repair Complete**

STEP 9. Identify state of FleetguardFIT™ sensor extension harnesses and FMS.

STEP 9A. Sensor breakout harness and extension harness condition.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Carefully inspect sensor breakout harness and extension harness	Are electrical harnesses in good condition with no damage, i.e. no tears? **Yes**	**9B**
	Are electrical harnesses in good condition with no damage, i.e. no tears? **No** **Repair:** Replace or repair harness	**Repair Complete**

STEP 9B. Sensor breakout harness and extension harness connection.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Carefully inspect sensor breakout harness and extension harness	Are electrical harnesses connected properly to FMS, breakout harness, and each sensor? **Yes**	**9C**
	Are electrical harnesses connected properly to FMS, breakout harness, and each sensor? **No** **Repair:** Connect wire(s) that are not connected	**Repair Complete**

STEP 9C. Sensor breakout harness and extension harness continuity.

Conditions : While equipment is off		
Action	**Specification/Repair**	**Next Step**
Measure continuity from end to end for breakout harness and each extension harness	Was continuity confirmed? **Yes** **Repair:** Replace sensor	**9D**
	Was continuity confirmed? **No** **Repair:** Fix or replace harnesses	**Repair complete**

STEP 9D. FMS module replacement.

Conditions : While equipment is off		
Specification/Repair	**Action**	**Next Step**
Is data available and within expected ranges? **Yes**	Log on to telematics portal and check faulty equipment for FleetguardFIT™ data	**Repair Complete**
Is data available and within expected ranges? **No** **Repair:** Replace FMS module		**Contact:** **FIT.Support@cummins.com**

Document History

Date	Details
2018-11-8	Module Created